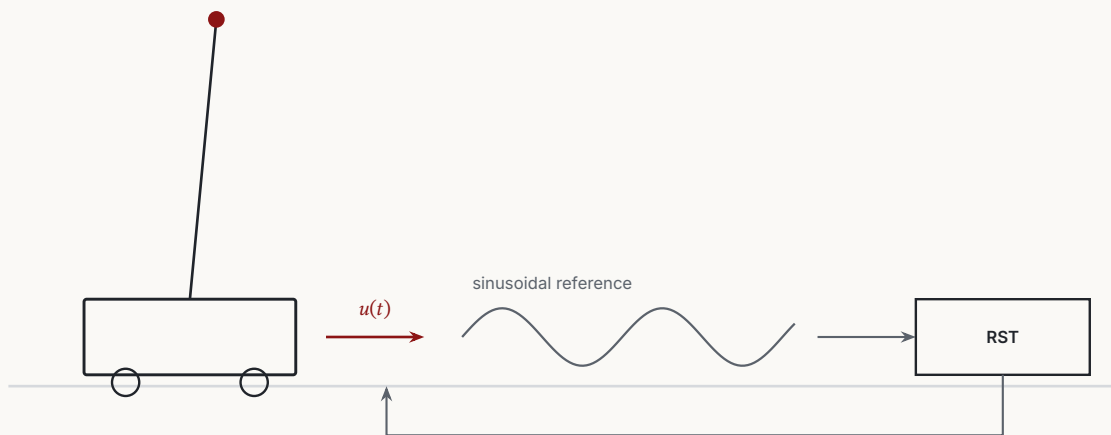


CONTROL SYSTEMS ENGINEERING

RST Polynomial Control of an Inverted Pendulum

Sinusoidal reference tracking, controller optimization,
and parametric robustness assessment



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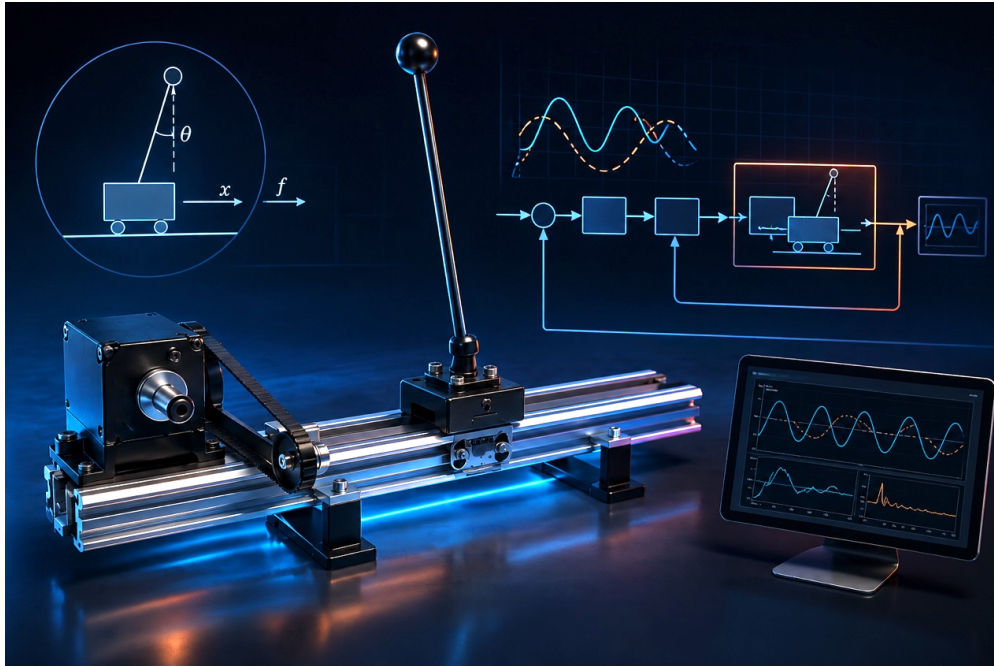
Abstract

The inverted pendulum is a canonical benchmark for the analysis of unstable, underactuated, and strongly coupled dynamical systems. This report develops a complete polynomial-control workflow for sinusoidal reference tracking. Starting from the nonlinear Lagrange equations, the plant is linearized about the upright equilibrium, decomposed into cart and pendulum subsystems, and reduced to a second-order unstable model after a justified time-scale separation. A two-degree-of-freedom RST controller is then synthesized through an internal-model constraint and a Bézout polynomial equation. The denominator polynomial is assigned by pole placement, while the feed-forward polynomial is selected to guarantee both sinusoidal and step tracking.

The initial design is evaluated in the frequency and time domains. Because the open-loop plant contains one unstable pole, the generalized Nyquist criterion is used in addition to conventional Bode indicators. The controller is subsequently retuned through the pole-location parameters β and γ . The selected design, $\beta = 0.5$ and $\gamma = 30$, yields a reported phase margin of 42.8° , maintains the control signal within ± 5 V, rejects the injected disturbance in approximately 1.55 s, and reduces the peak pendulum-angle excursion during disturbance rejection from 17.56° to 15.05° . Robustness is assessed for pendulum lengths $L \in \{0.16, 0.23, 0.28, 0.35\}$ m. Although the shortest pendulum drives the response toward the limits of the small-angle linearization, the cart-position loop remains stable and accurately regulated for all tested cases.

KEYWORDS inverted pendulum; RST polynomial controller; internal-model principle; sinusoidal tracking; pole placement; Bézout equation; generalized Nyquist criterion; robust control.

Executive Summary



Conceptual representation of the cart-pendulum platform, the feedback architecture, and the sinusoidal tracking objective.

ENGINEERING PROBLEM

Stabilize an open-loop unstable inverted pendulum while tracking a sinusoidal position reference and respecting a ± 5 V actuator limit.

CONTROL ARCHITECTURE

Two-degree-of-freedom RST polynomial controller with an integrator and a sinusoidal internal model embedded in $S(s)$.

SYNTHESIS METHOD

Lagrange modeling, linearization, time-scale separation, Bézout equation, closed-loop pole assignment, and MATLAB/Simulink verification.

OPTIMIZED DESIGN

$\beta = 0.5$, $\gamma = 30$, reported phase margin 42.8° , disturbance rejection approximately 1.55 s, and no simulated actuator saturation.

ROBUSTNESS SCOPE

Fixed-controller assessment for pendulum lengths from 0.16 m to 0.35 m; stable cart-position regulation is retained in all reported cases.

The equations, simulation outputs, and numerical values are preserved from the original technical project. The English prose, document architecture, captions, and scientific discussion have been comprehensively edited for clarity and academic presentation.

Nomenclature

M, m	Cart mass and pendulum mass, respectively.
l or L	Pendulum length.
x, θ	Cart displacement and pendulum angular displacement.
f, u	Applied force and controller command.
r, y, z	Reference, plant output, and measured output.
d_1, d_2, b	Input disturbance, output disturbance, and measurement noise.
$A(s), B(s)$	Denominator and numerator polynomials of the plant.
$R(s), S(s), T(s)$	Feedback, sensitivity-shaping, and feedforward polynomials of the RST controller.
$A_{BF}(s)$	Desired closed-loop characteristic polynomial.
ω_0	Sinusoidal-reference angular frequency.
ω_{0c}, ω_{0b}	Cart and pendulum natural angular frequencies.
ξ	Cart-loop damping ratio.
$\alpha = \sqrt{g/l}$	Unstable/stable pole magnitude of the reduced pendulum model.
β, γ	Dominant and fast closed-loop pole-placement parameters.
G_m, P_m, M_m	Gain margin, phase margin, and modulus margin.

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Introduction

The inverted pendulum combines three features that make it an enduring benchmark in control engineering: an open-loop unstable equilibrium, nonlinear coupling between translational and rotational motion, and a strict dependence of performance on actuator authority and model fidelity. Stabilization alone is therefore not sufficient. A practically useful controller must also track commands, reject disturbances, preserve adequate robustness margins, and avoid excessive control effort.

The objective of this project is to design and evaluate a continuous-time RST polynomial controller for an inverted pendulum mounted on a motorized cart. The commanded output is sinusoidal; consequently, the controller must include an internal model of the reference frequency. The design is carried out in a two-degree-of-freedom framework: $R(s)$ and $S(s)$ determine closed-loop stability and disturbance rejection, whereas $T(s)$ shapes reference tracking. The controller is derived analytically from a Bézout equation and implemented in MATLAB/Simulink.

The work proceeds in five stages. First, the steady-state tracking conditions and polynomial-degree constraints are established. Second, the cart-pendulum system is modeled from the Lagrange equations and experimentally identified. Third, a reduced second-order unstable transfer function is obtained by exploiting the separation between the fast cart dynamics and the slower pendulum dynamics. Fourth, the RST polynomials are synthesized by pole placement and tested in both frequency and time domains. Finally, the design is retuned and subjected to a parametric robustness study over four pendulum lengths.

The principal design requirements are:

- asymptotically accurate sinusoidal-reference tracking;
- closed-loop stability despite one open-loop unstable pole;
- rapid rejection of an additive disturbance;
- command magnitude bounded by ± 5 V;
- acceptable angular excursions within the range where the linearized model remains meaningful;
- robustness with respect to substantial variations in pendulum length.

RST Framework for Sinusoidal Tracking

WORK PACKAGE 1 RST structure for a sinusoidal input

2.1 Steady-state condition

For a sinusoidal input

$$e(t) = A \sin(\omega_0 t),$$

the steady-state output of a linear system $G(s)$ is

$$s(t) = A |G(i\omega_0)| \sin(\omega_0 t + \arg[G(i\omega_0)]).$$

To obtain a zero steady-state response at the forcing frequency, the frequency response must satisfy

$$|G(i\omega_0)| = 0.$$

2.2 Three-polynomial controller architecture

An RST controller is constructed from three polynomials, $R(s)$, $S(s)$, and $T(s)$. The plant is denoted

$$P(s) = \frac{B(s)}{A(s)}.$$

The signals in the preceding diagram are interpreted as follows:

- u : control input;
- y : plant output;

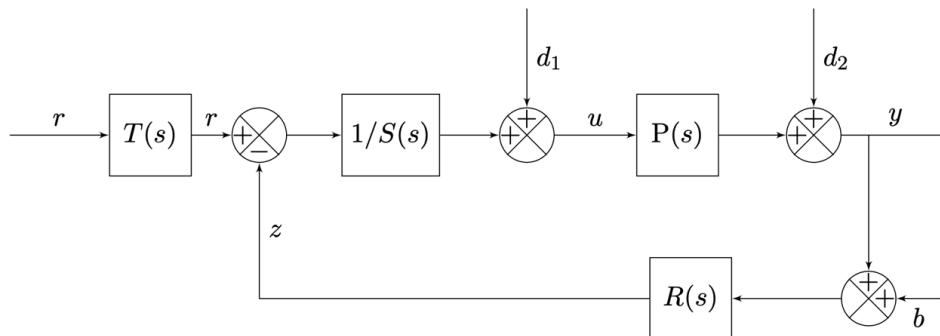


Figure 2.1: Closed-loop RST control architecture.

- z : measured output;
- r : reference command;
- b : measurement noise;
- d_1 : additive disturbance at the plant input;
- d_2 : additive disturbance at the plant output.

The reference-to-output transfer relation is

$$\hat{y}(s) = \frac{B(s)T(s)}{A(s)S(s) + B(s)R(s)} \hat{r}(s),$$

and the tracking error is

$$\hat{e}(s) = \hat{y}(s) - \hat{r}(s).$$

Therefore,

$$\hat{e}(s) = \frac{B(s)T(s) - A(s)S(s) - B(s)R(s)}{A(s)S(s) + B(s)R(s)} \hat{r}(s).$$

For $r(t) = A \sin(\omega_0 t)$, zero steady-state error requires

$$|B(i\omega_0)[T(i\omega_0) - R(i\omega_0)] - A(i\omega_0)S(i\omega_0)| = 0.$$

Since $A(i\omega_0) \neq 0$ and $B(i\omega_0) \neq 0$, the two design conditions are

$$T(i\omega_0) = R(i\omega_0) \quad \wedge \quad S(i\omega_0) = 0.$$

The first condition is enforced when $T(s)$ is selected. The second requires $S(s)$ to contain the conjugate pair $\pm i\omega_0$:

$$S(s) = (s^2 + \omega_0^2) F(s).$$

For simultaneous rejection of constant disturbances, an integrator is also introduced in $S(s)$.

2.3 Polynomial degree and solvability

Let

$$\deg R = k, \quad \delta_0 = \text{relative degree}.$$

The general feedback polynomial is

$$R(s) = r_0 s^k + r_1 s^{k-1} + r_2 s^{k-2} + \dots + r_k.$$

With the integrator and sinusoidal internal model embedded in $S(s)$,

$$S(s) = s(s^2 + \omega_0^2) (s^{k+\delta_0-3} + \sigma_1 s^{k+\delta_0-4} + \dots + \sigma_{k+\delta_0-3}).$$

The closed-loop characteristic polynomial is

$$A_{BF}(s) = A(s)S(s) + B(s)R(s),$$

whose degree is

$$\deg A_{BF} = n + k + \delta_0.$$

The number of unknown coefficients in R and S is

$$\text{card}(\mathcal{R} \cup \mathcal{S}) = 2k + \delta_0 - 2.$$

Thus, the design problem contains $n + k + \delta_0$ scalar equations and $2k + \delta_0 - 2$ unknowns. An exactly determined coefficient-identification problem requires

$$n + k + \delta_0 = 2k + \delta_0 - 2,$$

which gives

$$k = n + 2.$$

This degree relation is used later for the reduced second-order pendulum model.

The cart-pendulum dynamics are obtained from the Lagrange equations:

Linearization about the upright configuration $\theta = 0$ yields

WORK PACKAGE 2 System analysis and transfer-function identification

$$\begin{cases} f(s) - (M + m)s^2 X(s) = ml\theta(s), \\ ls^2\theta(s) = s^2 X(s) - g\theta(s). \end{cases}$$
$$\begin{cases} X(s) = \frac{1}{(M+m)s^2}f(s) - \frac{ml}{(M+m)s^2}\theta(s), \\ \theta(s) = \frac{s^2}{g-ls^2}X(s). \end{cases}$$

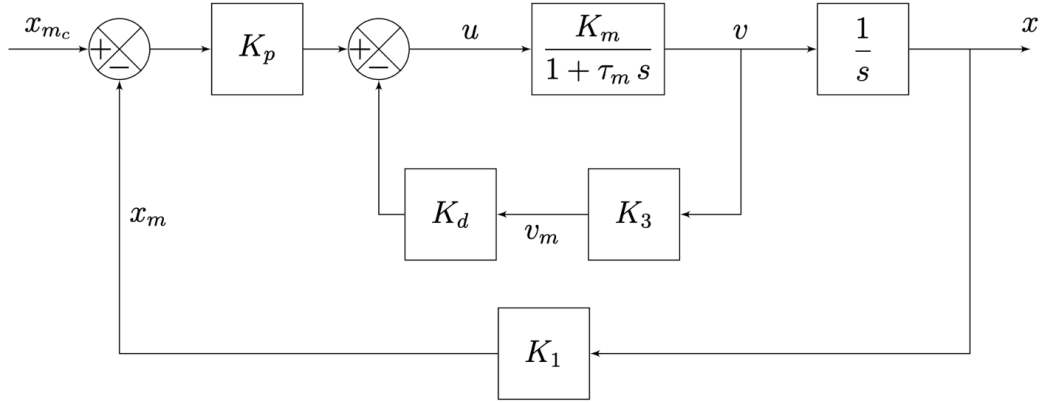



Figure 3.2: Modified cart-position servo-control diagram.

Identification with the block diagram gives

$$G_1(s) = \frac{1}{(M+m)s^2}, \quad G_2(s) = -\frac{ml}{M+m}, \quad G_3(s) = \frac{s^2}{g-ls^2}.$$

3.2 Closed-loop transfer function of the cart-position servo

WORK PACKAGE 3 Closed-loop cart-position transfer function

The command, cart velocity, and measured position satisfy

$$u(s) = K_p[x_{mc}(s) - x_m(s)] - K_d K_3 v(s),$$

$$v(s) = \frac{K_m}{1 + \tau_m s} u(s),$$

$$x(s) = \frac{v(s)}{s}, \quad x_m(s) = K_1 x(s).$$

After substitution and collection of terms,

$$\frac{x_m(s)}{x_{mc}(s)} = \frac{K_m K_p K_1}{\tau_m s^2 + (1 + K_m K_d K_3)s + K_m K_p K_1}.$$

Hence,

$$H_1(s) = \frac{1}{\frac{\tau_m}{K_m K_p K_1} s^2 + \frac{1 + K_m K_d K_3}{K_m K_p K_1} s + 1}.$$

In canonical second-order form,

$$H_1(s) = \frac{1}{\frac{s^2}{\omega_{0c}^2} + \frac{2\xi}{\omega_{0c}} s + 1},$$

with

$$\omega_{0c} = \sqrt{\frac{K_m K_p K_1}{\tau_m}},$$

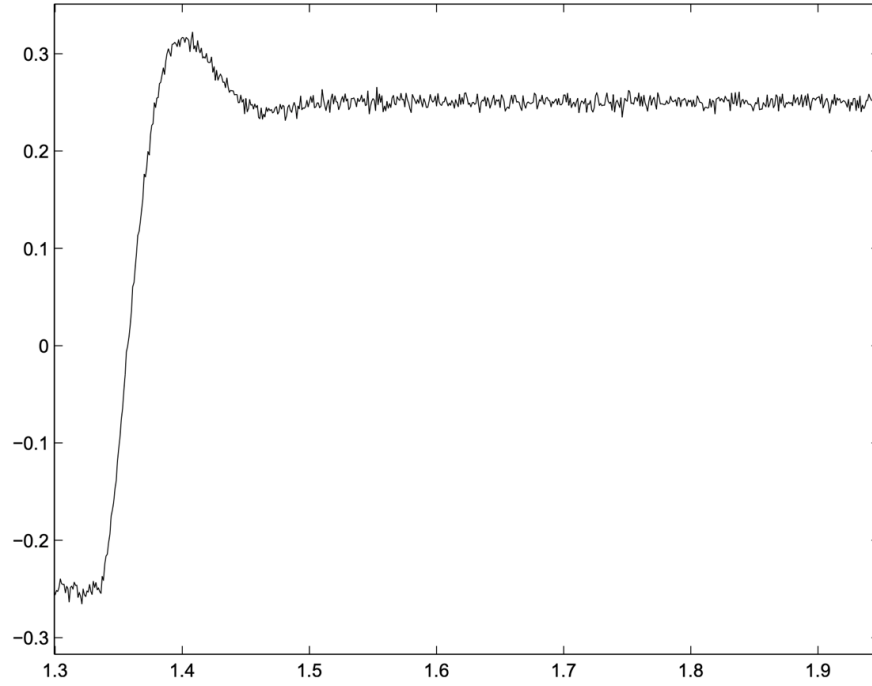


Figure 3.3: Internal cart-position servo: closed-loop identification test.

and

$$\frac{2\xi}{\omega_{0c}} = \frac{1 + K_m K_d K_3}{K_m K_p K_1}, \quad \xi = \frac{1 + K_m K_d K_3}{2K_m K_p K_1} \sqrt{\frac{K_m K_p K_1}{\tau_m}}.$$

3.3 Experimental identification of the cart loop

WORK PACKAGE 4 Cart-position servo identification

A square-wave command of frequency $f = 1.5$ Hz and peak-to-peak amplitude $A_c = 0.5$ V was applied, corresponding to $x_{mc} = \pm 0.25$ V. The measured response $x_m(t)$ is shown below.

The final-value theorem gives

$$G_{\text{static}} = \frac{\lim_{t \rightarrow +\infty} x_m(t)}{x_{mc}} = \frac{0.25}{0.25}, \quad \boxed{G_{\text{static}} = 1}.$$

The relative overshoots are

$$D_{\%} = \frac{|y_{\text{max/min}} - y_{\infty}|}{y_{\infty}} \times 100,$$

$$d_{1\%} = \frac{0.31 - 0.25}{0.25} \times 100 = 24\%, \quad d_{2\%} = \frac{|0.24 - 0.25|}{0.25} \times 100 = 4\%.$$

For a second-order response,

$$\frac{d_2}{d_1} = \exp\left(-\frac{\pi\xi}{\sqrt{1-\xi^2}}\right),$$

so that

$$\xi = \frac{-\ln(d_{2\%}/d_{1\%})}{\sqrt{\pi^2 + \ln^2(d_{2\%}/d_{1\%})}}, \quad \omega_{0c} = \frac{\pi}{\sqrt{1-\xi^2}} \frac{1}{t_2 - t_1}.$$

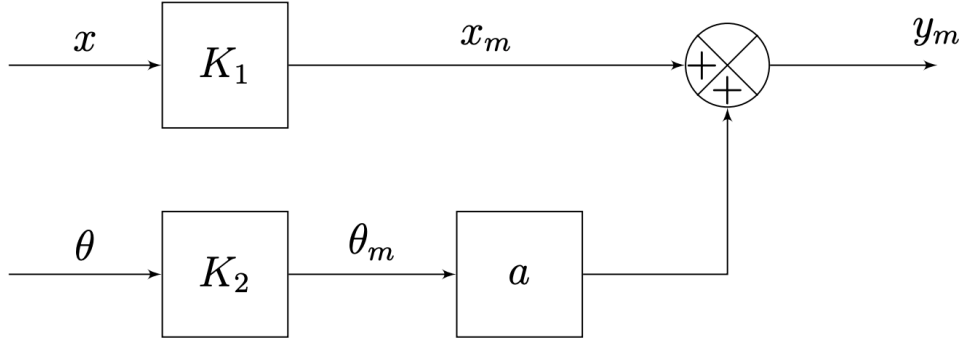


Figure 3.4: Measurement architecture for cart displacement and pendulum angle.

Using $t_1 = 1.41$ s and $t_2 = 1.48$ s,

$$\boxed{\xi \approx 0.495}, \quad \boxed{\omega_{0c} \approx 51.66 \text{ rad s}^{-1}}.$$

3.4 Position and angle measurement system

WORK PACKAGE 5 Pendulum-position sensing model

The available measurement combines cart position and pendulum angle.

For small angles,

$$y(s) = x(s) + l\theta(s).$$

From the measurement diagram,

$$y_m(s) = K_1 x(s) + aK_2 \theta(s).$$

Let $y_m(s) = \lambda y(s)$. Coefficient matching gives

$$\begin{cases} \lambda = K_1, \\ \lambda l = K_2 a, \end{cases}$$

and therefore

$$\boxed{\lambda = K_1}, \quad \boxed{a = \frac{K_1}{K_2} l}.$$

3.5 Pendulum dynamics and time-scale separation

WORK PACKAGE 6 Pendulum dynamics

The pendulum transfer function is

$$G_3(s) = \frac{s^2}{g - ls^2} = \frac{s^2/l}{g/l - s^2}.$$

Its poles are $\pm\sqrt{g/l}$, so the undamped natural frequency is

$$\omega_{0b} = \sqrt{\frac{g}{l}}.$$

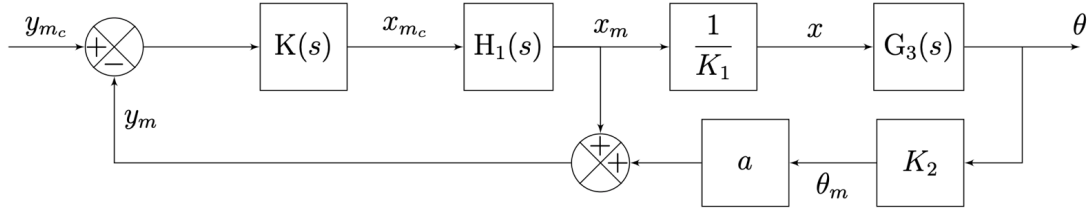


Figure 3.5: Reduced control architecture after neglecting the coupling block $G_2(s)$.

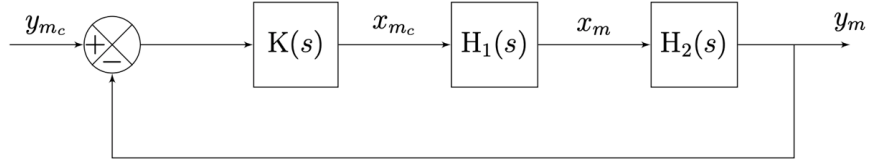


Figure 3.6: Equivalent cascade representation of the reduced control loop.

For $g = 9.81 \text{ m s}^{-2}$, the values are:

Table 3.1: Pendulum natural frequency as a function of length.

$l \text{ (m)}$	$\omega_{0b} \text{ (rad s}^{-1}\text{)}$
0.16	7.83
0.23	6.53
0.28	5.92
0.35	5.29

Since $\omega_{0b} < \omega_{0c}$, the pendulum is the slow subsystem and the cart-position loop is comparatively fast. Over the frequency range relevant to the pendulum, the coupling represented by $G_2(s)$ is neglected. The cart subsystem is then replaced by the equivalent closed-loop transfer function $H_1(s)$.

3.6 Equivalent reduced plant

WORK PACKAGE 7 Equivalent control model

From the sensing relation,

$$y_m = K_1 x_m + a K_2 \theta_m.$$

Using $a = K_1 l / K_2$,

$$y_m = K_1 (x + l \theta) = K_1 y.$$

The cart displacement and pendulum angle may be expressed in terms of x_{mc} as

$$x_m = H_1(s) x_{mc}, \quad x = \frac{H_1(s) x_{mc}}{K_1},$$

$$\theta = G_3(s) x = G_3(s) \frac{H_1(s) x_{mc}}{K_1}.$$

Substituting into y_m gives

$$y_m = H_1(s)x_{mc}[1 + lG_3(s)].$$

Moreover,

$$1 + lG_3(s) = 1 + l \frac{s^2}{g - ls^2} = \frac{g}{g - ls^2}.$$

With

$$\tau = \sqrt{\frac{l}{g}} = \frac{1}{\alpha},$$

this term becomes

$$\frac{g}{g - ls^2} = \frac{1}{1 - \tau^2 s^2} = \frac{1}{(1 - \tau s)(1 + \tau s)}.$$

Consequently,

$$\frac{y_m}{x_{mc}} = H_1(s)H_2(s),$$

where

$$H_2(s) = \frac{1}{(1 + \tau s)(1 - \tau s)}, \quad \tau = \sqrt{\frac{l}{g}} = \frac{1}{\alpha}.$$

RST Controller Synthesis

4.1 Reduced model for controller design

Tracking a sinusoid increases the controller order relative to a standard RST design because the internal model must contain the poles $\pm i\omega_0$. To keep the synthesis tractable, the fast cart loop $H_1(s)$ is approximated by its static gain, equal to unity. The design plant is therefore the second-order unstable block $H_2(s)$.

The reduced plant is

$$H_2(s) = \frac{1}{(1 + \tau s)(1 - \tau s)} = \frac{-\alpha^2}{s^2 - \alpha^2} = \frac{B(s)}{A(s)},$$

with

$$\boxed{B(s) = -\alpha^2}, \quad \boxed{A(s) = s^2 - \alpha^2}.$$

4.2 Structures of $R(s)$ and $S(s)$

WORK PACKAGE 8 Identification of the $R(s)$ and $S(s)$ polynomials

Here $n = \deg A = 2$. From $k = n + 2$, one obtains

$$k = 4, \quad \deg A_{BF} = 2n + 2 + \delta_0.$$

Taking $\delta_0 = 0$ gives a sixth-order closed-loop characteristic polynomial. The feedback polynomial is

$$\boxed{R(s) = r_0 s^4 + r_1 s^3 + r_2 s^2 + r_3 s + r_4}.$$

The sensitivity polynomial must contain an integrator and the sinusoidal internal model:

$$\boxed{S(s) = s(s^2 + \omega_0^2)(s + \sigma_1)}.$$

4.3 Closed-loop pole placement

The minimum-speed line is placed at $-\beta$, while the additional fast pole is placed at $-\gamma$, with $(\beta, \gamma) \in (\mathbb{R}^+)^2$. The plant contributes the poles $\{-\alpha, +\alpha\}$; the internal model contributes $\{0, -i\omega_0, +i\omega_0\}$; and the remaining design freedom is represented by $-\gamma$. The unstable open-loop pole is reassigned to the stable half-plane, yielding the

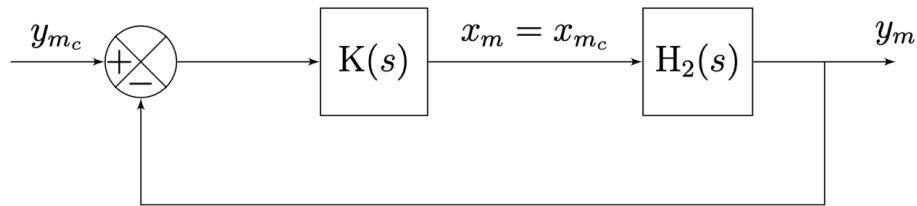


Figure 4.1: Simplified closed-loop model used for RST synthesis.

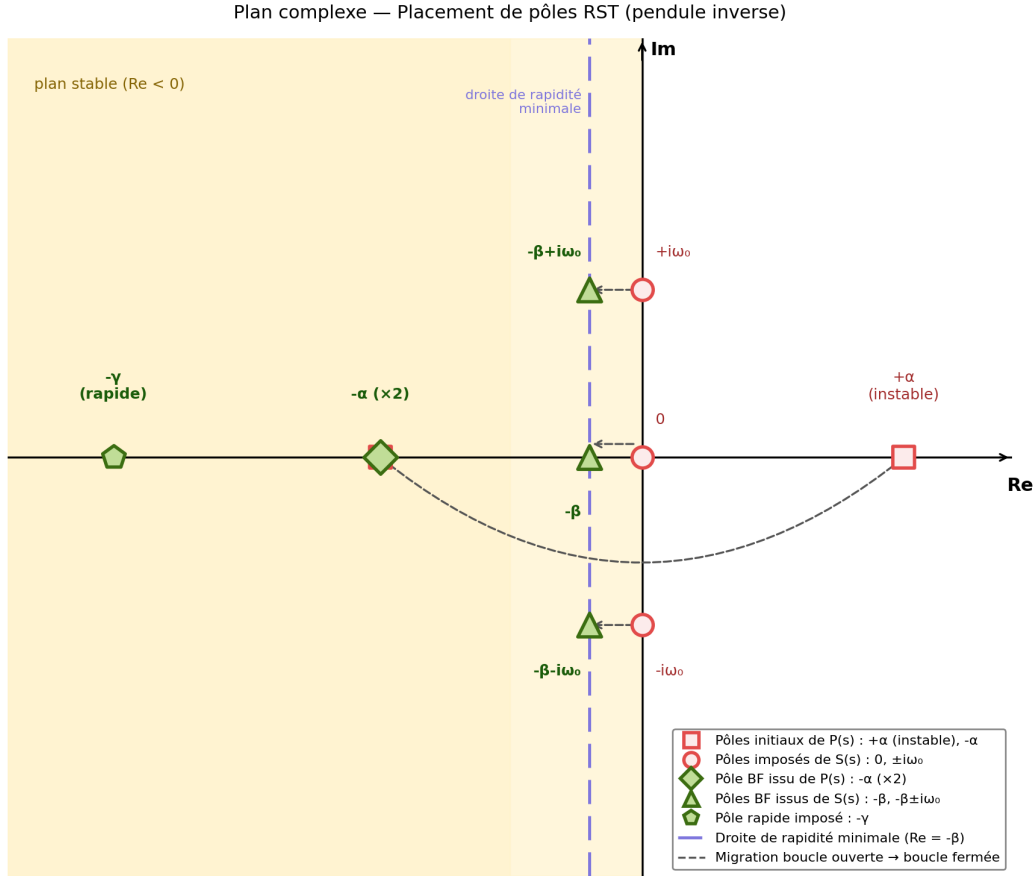


Figure 4.2: Closed-loop pole-placement strategy in the complex plane (generated with Python/Matplotlib).

desired set

$$\mathcal{P}_{BF} = \{-\gamma, -\alpha, -\alpha, -\beta, -\beta - i\omega_0, -\beta + i\omega_0\}.$$

The desired characteristic polynomial is therefore

$$A_{BF}(s) = (s + \gamma)(s + \alpha)^2(s + \beta)(s + \beta + i\omega_0)(s + \beta - i\omega_0).$$

The initial test values are

$$\beta = 1, \quad \gamma = 10, \quad L = 0.35 \text{ m.}$$

The last value provides a comparatively slow pendulum and is used as the baseline before robustness is tested over the full length range.

Writing

$$A_{BF}(s) = s^6 + c_1 s^5 + c_2 s^4 + c_3 s^3 + c_4 s^2 + c_5 s + c_6,$$

MATLAB returns

$$\begin{aligned} c_1 &= 23.5884, & c_2 &= 199.6778, & c_3 &= 766.3771, \\ c_4 &= 1417.6844, & c_5 &= 1388.9681, & c_6 &= 560.5714. \end{aligned}$$

```

Command Window
>> TP2_RST_Pendule_Dev_KUMAR
c1=23.5884 c2=199.6778 c3=766.3771 c4=1417.6844 c5=1388.9681 c6=560.5714
>>

```

Figure 4.3: MATLAB output for the coefficients of $A_{BF}(s)$.

```

Command Window
Sigma1=23.5884
r0=-8.0884 r1=-50.0895 r2=-51.5800 r3=-73.1438 r4=-20.0000
>>

```

Figure 4.4: MATLAB solution of the Bézout equation for $R(s)$ and $S(s)$.

4.4 Bézout equation and computation of $R(s)$ and $S(s)$

The coefficient-identification problem is governed by the Bézout equation

$$A_{BF}(s) = A(s)S(s) + B(s)R(s).$$

Substitution gives

$$A_{BF}(s) = (s^2 - \alpha^2)s(s^2 + \omega_0^2)(s + \sigma_1) - \alpha^2(r_0s^4 + r_1s^3 + r_2s^2 + r_3s + r_4).$$

Matching coefficients with $s^6 + c_1s^5 + \dots + c_6$ yields

$$\begin{cases} \sigma_1 = c_1, \\ r_0 = \frac{\omega_0^2 - \alpha^2 - c_2}{\alpha^2}, \\ r_1 = \frac{c_1(\omega_0^2 - \alpha^2) - c_3}{\alpha^2}, \\ r_2 = -\frac{\alpha^2\omega_0^2 + c_4}{\alpha^2}, \\ r_3 = -\frac{\alpha^2c_1\omega_0^2 + c_5}{\alpha^2}, \\ r_4 = -\frac{c_6}{\alpha^2}. \end{cases}$$

For the initial design,

$$\begin{aligned} \sigma_1 &= 23.5884, \\ r_0 &= -8.0884, \quad r_1 = -50.0895, \\ r_2 &= 51.58, \quad r_3 = -73.1438, \quad r_4 = -20. \end{aligned}$$

4.5 Identification of the feedforward polynomial $T(s)$

WORK PACKAGE 9 Identification of $T(s)$

The degree constraint is

$$\deg T(s) \leq \deg S(s) = 4.$$

```

Command Window
t0=-1.7912  t1=-2.3064  t2=-2.0000
>>

```

Figure 4.5: MATLAB output for the coefficients of $T(s)$.

To compensate the poles $-\gamma$ and $-\beta$, the feedforward polynomial is selected as

$$T(s) = (s + \gamma)(s + \beta)(t_0 s^2 + t_1 s + t_2).$$

The tracking requirements are

$$T(i\omega_0) = R(i\omega_0) \quad \text{and} \quad T(0) = R(0).$$

With $\omega_0 = 1 \text{ rad s}^{-1}$,

$$\begin{cases} (i + \gamma)(i + \beta)(-t_0 + it_1 + t_2) = r_0 - ir_1 - r_2 + ir_3 + r_4, \\ \gamma\beta t_2 = r_4. \end{cases}$$

Separating real and imaginary parts leads to

$$\begin{pmatrix} 1 - \gamma\beta & -(\gamma + \beta) \\ -(\gamma + \beta) & \gamma\beta - 1 \end{pmatrix} \begin{pmatrix} t_0 \\ t_1 \end{pmatrix} = \begin{pmatrix} r_0 - r_2 + r_4 - (\gamma\beta - 1)t_2 \\ r_3 - r_1 - (\gamma + \beta)t_2 \end{pmatrix},$$

with

$$t_2 = \frac{r_4}{\gamma\beta}.$$

The computed coefficients are

$$t_0 = -1.7912, \quad t_1 = -2.3064, \quad t_2 = -2.$$

Initial Controller Evaluation

WORK PACKAGE 10 Computation and validation of the initial RST controller

5.1 Frequency-domain analysis

The open-loop transfer function associated with the R - S part of the controller is evaluated in MATLAB. In addition to gain and phase margins, the modulus margin is computed as the minimum distance between the Nyquist locus and the critical point -1 :

$$M_m = \min_{\omega} |1 + L(i\omega)|.$$

The low-frequency magnitude slope is approximately -20 dB/decade, which is consistent with the integrator embedded in $S(s)$ and therefore with rejection of constant disturbances. A pronounced resonance appears at $\omega = 1$ rad s^{-1} , exactly at the sinusoidal internal-model frequency imposed by the factor $s^2 + \omega_0^2$. The corresponding phase variation reflects the pair of imaginary-axis poles introduced in the open-loop controller structure.

MATLAB reports a gain margin of -3.82 dB at $\omega = 2.77$ rad s^{-1} and a phase margin of approximately 30° at $\omega = 7.44$ rad s^{-1} . These values must be interpreted cautiously because the plant contains the open-loop unstable pole $+\alpha$. Conventional Bode-margin interpretations assume open-loop stability and are therefore not, by themselves, conclusive here. Closed-loop stability is instead corroborated by the roots of $A_{BF}(s)$, all of which have negative real parts, and by the convergent time response.

5.2 Generalized Nyquist verification

The generalized Nyquist criterion is applicable to an open-loop unstable plant. Since the reduced process has one right-half-plane pole, $P = 1$, closed-loop stability requires one counter-clockwise encirclement of the critical point -1 . The reported locus crosses the real axis near -1.53 and encloses -1 in the required direction. Thus $N = P = 1$, and the closed loop is stable.

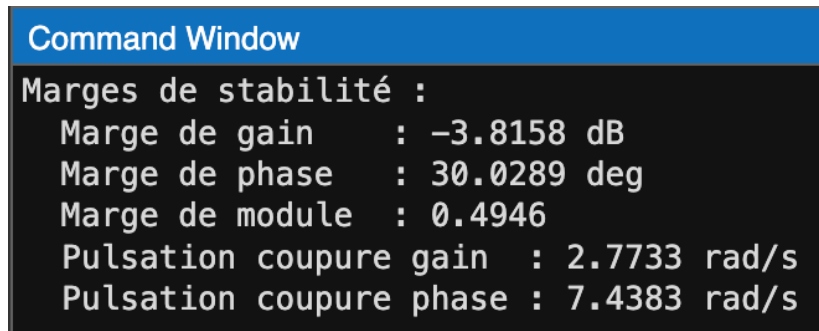


Figure 5.1: Reported stability indicators for the initial RST design.

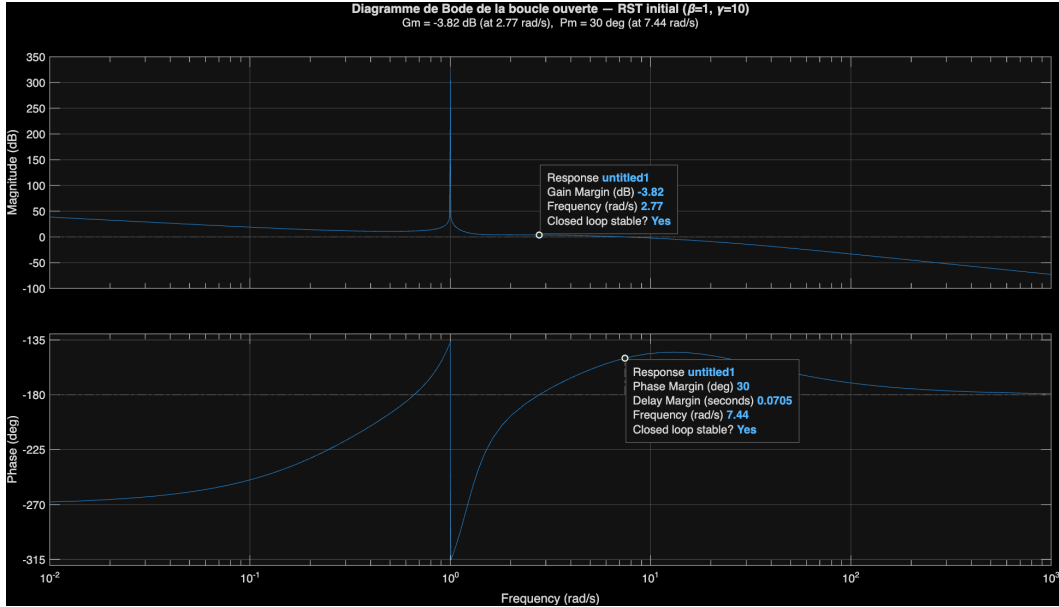


Figure 5.2: Open-loop Bode diagram with the initial RST controller.

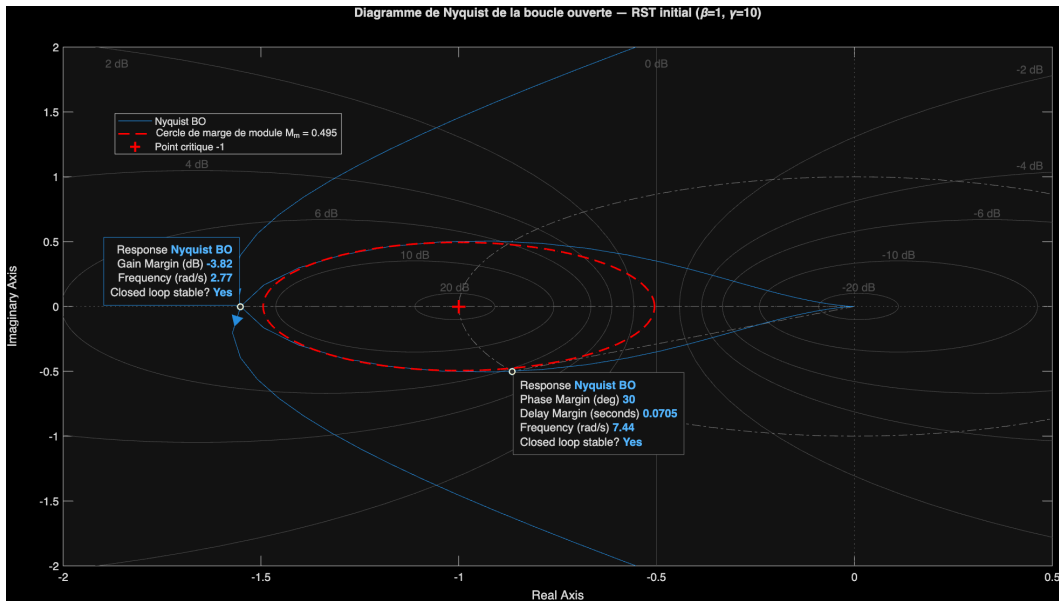


Figure 5.3: Nyquist diagram for the initial RST design.

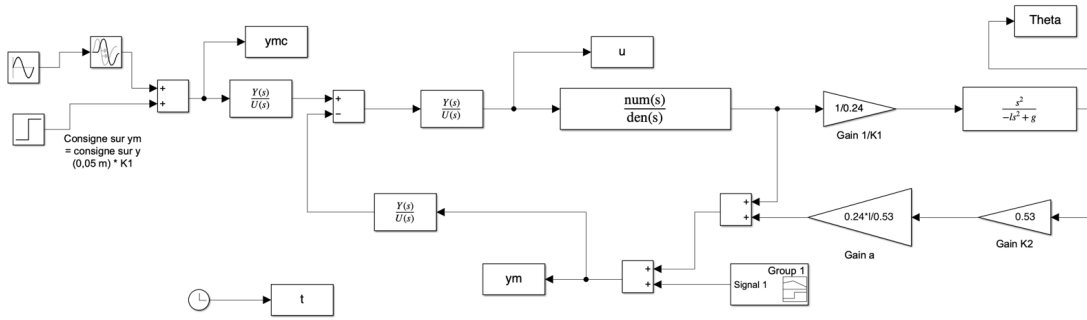


Figure 5.4: Simulink implementation of the RST-controlled inverted-pendulum model.

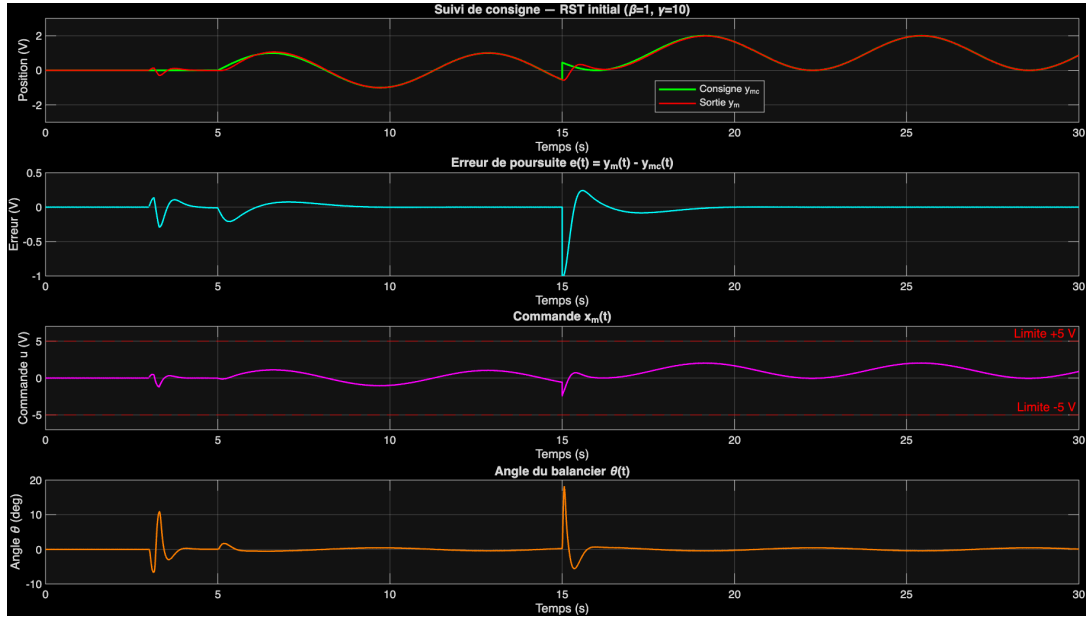


Figure 5.5: Closed-loop response with the initial RST controller.

The initial modulus margin is approximately

$$M_m \approx 0.49.$$

This value lies just below the commonly used robustness reference of 0.5, indicating that the Nyquist locus passes relatively close to the critical point. The result motivates retuning β and γ to increase robustness while preserving tracking performance and actuator feasibility.

5.3 Simulink implementation and time-domain response

The reduced block diagram is implemented in Simulink as shown below.

For the initial parameter set $\beta = 1$ and $\gamma = 10$, the reference and measured output are nearly indistinguishable over most of the simulation horizon.

The control command remains within the prescribed interval $[-5, 5]$ V. The controller also rejects the disturbance introduced at $t = 15$ s.

The measured disturbance-rejection response time is approximately 1.35 s. This confirms that the internal-model requirement does not prevent rapid recovery from a transient perturbation.

The angular response provides a physical measure of the feasibility of the linear design. A short calibration transient appears at the beginning of the simulation, partly because the analytical derivation was

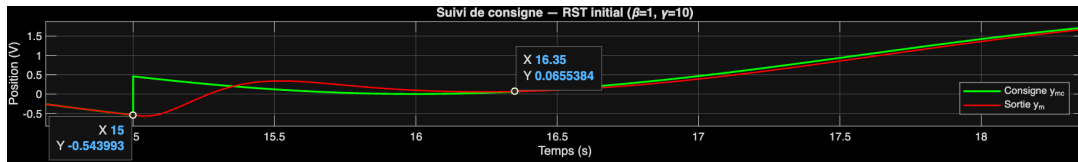


Figure 5.6: Detailed view of disturbance rejection with the initial controller.

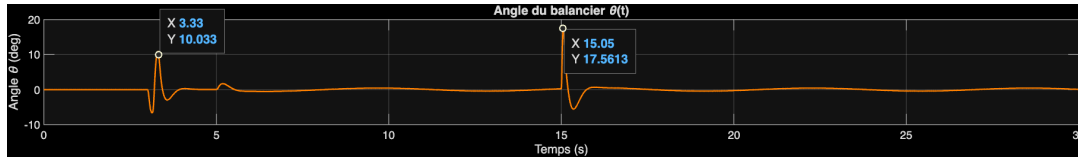


Figure 5.7: Pendulum-angle response with the initial RST controller.

performed under zero initial conditions. The peak angle remains below 20° in this baseline case, and the reported maximum excursion during the disturbance is 17.56° .

The limitation of the model must nevertheless be kept in view. The controller is derived from a linearization in which the nonlinear term $ml\dot{\theta}^2 \sin \theta$ is neglected relative to $ml\ddot{\theta} \cos \theta$. If the angular excursion becomes too large, the small-angle approximation degrades and the linear RST controller can no longer be expected to reproduce the true nonlinear dynamics accurately.

RST Controller Optimization

6.1 Retuning rationale

The initial controller already satisfies the tracking and actuator constraints, but its robustness margin is limited. Retuning therefore focuses on increasing the phase margin while preserving disturbance rejection, avoiding resonant closed-loop behavior, and maintaining the ± 5 V command bound.

The identified natural frequencies are

$$\omega_{0b} = 5.29 \text{ rad s}^{-1} \quad \text{and} \quad \omega_{0c} = 51.66 \text{ rad s}^{-1}$$

for the baseline pendulum length. The intended separation of time scales is expressed by

$$0 < \beta \ll \omega_{0b} \ll \gamma \ll \omega_{0c}.$$

This hierarchy prevents the dominant closed-loop dynamics from becoming faster than the pendulum itself and prevents the fast controller pole from forcing the cart beyond its useful bandwidth. Consequently,

$$\beta \in (0, 5.29), \quad \gamma \in (5.29, 51.66).$$

For robustness over all tested lengths, the lower pendulum frequency, $\min \omega_{0b} = 5.29 \text{ rad s}^{-1}$, is used as the conservative reference.

A dichotomic parameter search was performed. The selected values are

$$\boxed{\beta = 0.5}, \quad \boxed{\gamma = 30}.$$

They provide the best reported phase-margin improvement above the 40° target without command saturation.

6.2 Optimized frequency-domain response

The optimized design achieves a reported phase margin of

$$\boxed{P_m = 42.8^\circ},$$

which satisfies the imposed robustness requirement.

The generalized Nyquist criterion remains satisfied: the locus produces the counter-clockwise encirclement required by the single unstable open-loop pole $+\alpha$. The optimization therefore improves the frequency-domain robustness indicator without compromising closed-loop stability.

6.3 Optimized time-domain performance

The optimized response remains very close to the initial response in terms of reference tracking. No actuator saturation is observed, and the command remains within $[-5, 5]$ V.

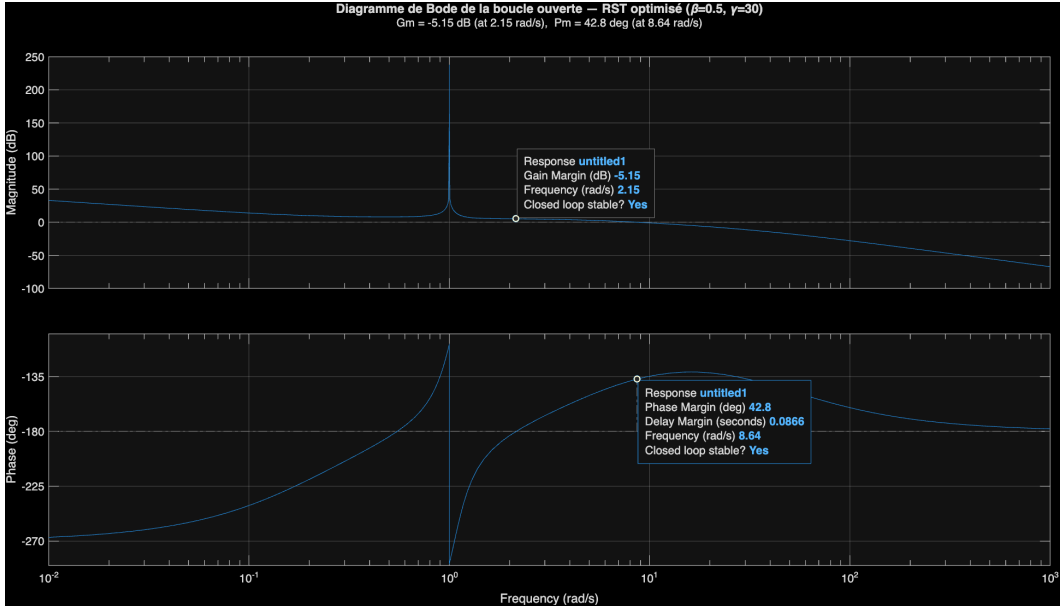


Figure 6.1: Open-loop Bode diagram with the optimized RST controller.

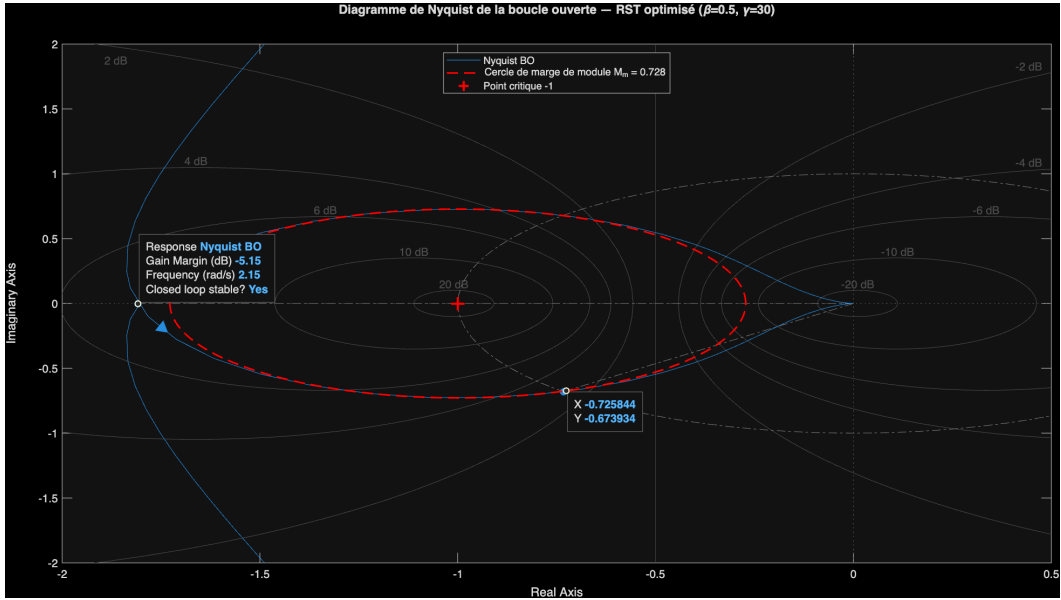


Figure 6.2: Nyquist diagram with the optimized RST controller.

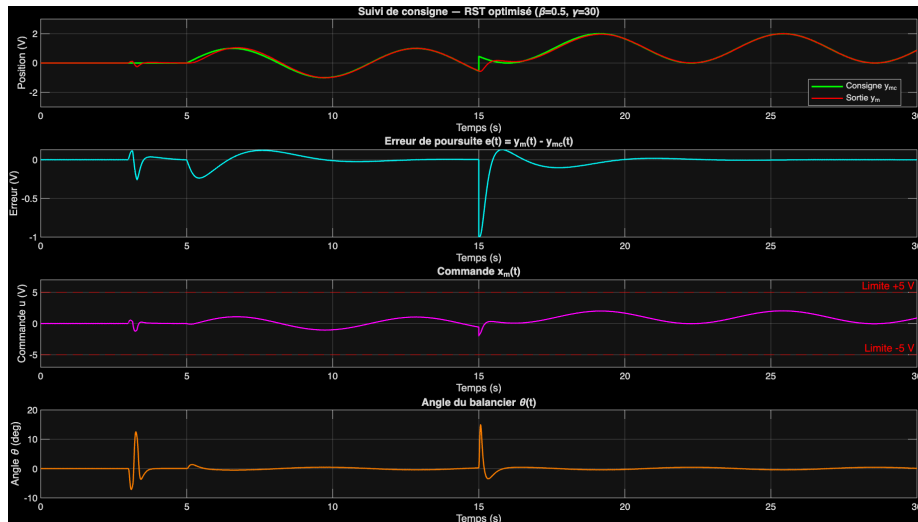


Figure 6.3: Time-domain performance of the optimized RST controller.

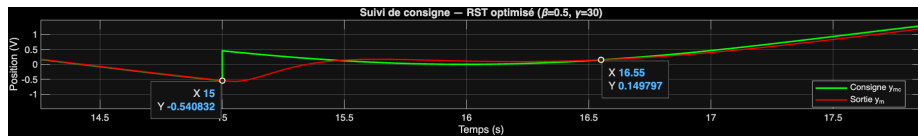


Figure 6.4: Disturbance-rejection detail for the optimized RST controller.

The disturbance-rejection response time is approximately 1.55 s. The small increase relative to the initial 1.35 s response is the expected cost of moving toward a more robust, less aggressive closed-loop design.

The peak angular excursion decreases from 17.56° to 15.05° . The optimized controller therefore preserves tracking speed while increasing the stability margin with respect to the linear-model validity range.

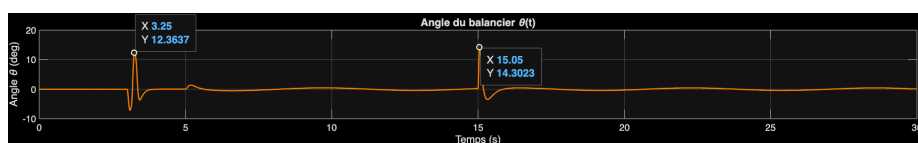


Figure 6.5: Pendulum-angle response with the optimized RST controller.

Parametric Robustness Study

7.1 Test protocol

The optimized controller is held fixed at

$$\beta = 0.5, \quad \gamma = 30,$$

while the pendulum length is varied over

$$L \in \{0.16, 0.23, 0.28, 0.35\} \text{ m.}$$

For each length, the unstable-pole magnitude is recomputed as $\alpha_k = \sqrt{g/L_k}$. The open-loop margins, modulus margin, maximum command, cart response, and pendulum angle are then evaluated without redesigning the controller.

The reported phase margin remains above 48° for the tested length variations. Although individual classical margins must still be interpreted in the context of an unstable open loop, the combination of generalized Nyquist verification and convergent time responses indicates that the fixed controller remains globally stable and robust over the investigated parameter set.

7.2 Reference tracking and actuator demand

The four output curves follow the same overall profile and remain close to the reference. No control-command saturation is observed, so the simulated actuator remains within its admissible operating range for every length.

7.3 Disturbance rejection

The disturbance is rejected in approximately 1.56 s for all four values of L . Cart-position regulation is therefore only weakly influenced by pendulum length. This is consistent with the assigned closed-loop polynomial: the length variation primarily modifies the repeated poles at $-\alpha$, whereas the dominant tracking and disturbance-

Command Window					
===== TEST DE ROBUSTESSE – Variation de L =====					
(Correcteur RST optimisé maintenu : beta=0.50, gamma=30.00)					
L (m)	alpha	Gm (dB)	Pm (deg)	Mm	u max (V)
0.16	7.8302	-5.8429	48.4734	0.7026	2.0142
0.23	6.5309	-5.5794	47.9730	0.7388	2.0213
0.28	5.9191	-5.3959	46.1501	0.7505	2.0265
0.35	5.2942	-5.1455	42.8365	0.7277	2.0336
>>					

Figure 7.1: Stability margins, tracking error, and control command for the four pendulum lengths.

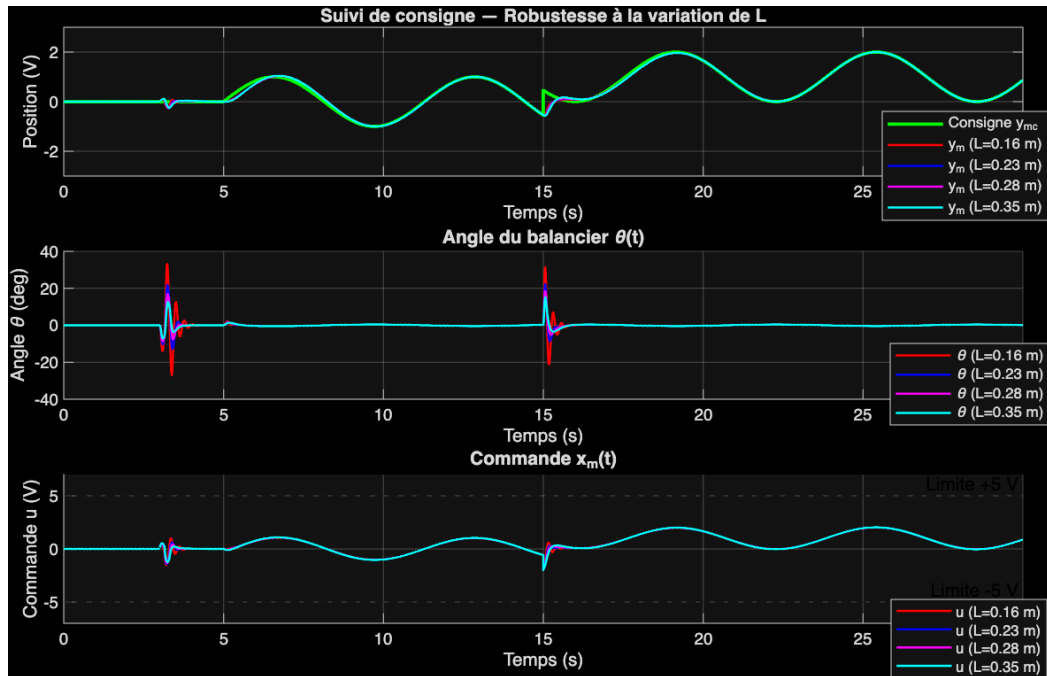


Figure 7.2: Optimized RST responses for varying pendulum length.

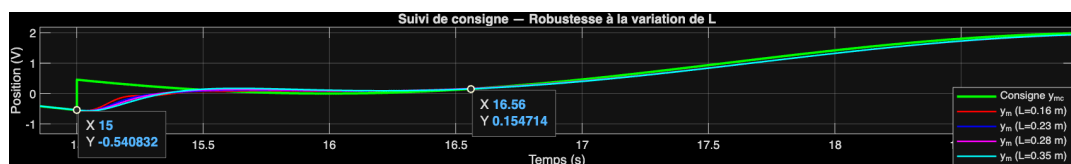


Figure 7.3: Disturbance-rejection comparison for the robustness study.

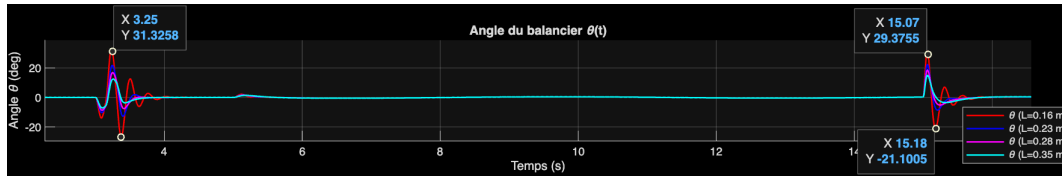


Figure 7.4: Pendulum-angle response for the optimized controller and varying length.

rejection dynamics are largely governed by the remaining assigned poles.

7.4 Angular response and validity of the linear model

The peak angular excursion increases as L decreases. A shorter pendulum has a larger natural frequency and a right-half-plane pole $+\alpha$ located farther from the imaginary axis. Its open-loop divergence is therefore faster and more difficult to arrest. For $L = 0.16$ m, the reported calibration transient reaches 31.33° . At this amplitude, the small-angle approximation is no longer especially accurate, and the response approaches the boundary of validity of the linear model.

Despite this limitation, the cart-position output remains stable and regulated for all four lengths. The result demonstrates meaningful parametric robustness of the optimized RST design, including cases that extend beyond the strict assumptions used during linear synthesis. It does not, however, constitute a guarantee of nonlinear global stability; that question requires validation on the full nonlinear model or on the experimental platform.

Conclusion

This study developed a complete RST polynomial-control design for an inverted pendulum required to track a sinusoidal reference. The nonlinear cart-pendulum equations were first established from Lagrange's formalism and linearized about the upright equilibrium. Experimental identification of the cart-position loop produced a unit static gain, a damping ratio $\xi \approx 0.495$, and a natural frequency $\omega_{0c} \approx 51.66 \text{ rad s}^{-1}$. Comparison with the pendulum frequencies justified a reduced second-order design model in which the fast cart dynamics were approximated by their static gain.

The sinusoidal tracking requirement was incorporated directly through the internal-model principle. The polynomial $S(s)$ contains both an integrator and the factor $s^2 + \omega_0^2$, while $T(s)$ satisfies $T(i\omega_0) = R(i\omega_0)$. The degree-counting argument led to $k = n + 2$, and the controller coefficients were obtained from the Bézout equation $A_{BF} = AS + BR$ after assigning a sixth-order closed-loop polynomial. This construction provides a transparent link between the desired pole set and the resulting controller.

The initial design achieved accurate tracking, respected the $\pm 5 \text{ V}$ actuator constraint, and rejected the imposed disturbance in approximately 1.35 s. Since the reduced plant contains one unstable open-loop pole, stability was verified with the generalized Nyquist criterion rather than inferred solely from conventional Bode margins. The initial modulus margin of approximately 0.49 motivated a retuning of the pole-placement parameters.

The optimized choice $\beta = 0.5$ and $\gamma = 30$ achieved a reported phase margin of 42.8° , maintained disturbance rejection at approximately 1.55 s, and reduced the disturbance-induced peak pendulum angle from 17.56° to 15.05° . The robustness study further showed stable cart-position tracking and unsaturated control for $L = 0.16, 0.23, 0.28$, and 0.35 m , with an almost invariant disturbance-rejection time near 1.56 s.

The main limitation is the linearization itself. For the shortest pendulum, the calibration angle reaches 31.33° , where the neglected nonlinear terms may no longer be insignificant. Future work should therefore extend the validation to the complete nonlinear model, include friction, sensor noise, actuator dynamics, saturation, delay, and parameter uncertainty, and compare the RST solution with gain-scheduled, H_∞ , or nonlinear control strategies. Experimental implementation would provide the decisive assessment of robustness under unmodeled dynamics and real measurement noise.

MATLAB Implementation Extracts

The following listings reproduce the computational steps used in the report. Comments, console messages, and plot labels have been translated into English; variable names and numerical operations are unchanged.

A.1 Closed-loop polynomial and initial coefficients

```

1  %% Closed-loop polynomial
2  l = 0.35;
3  g = 9.81;
4  alpha = sqrt(g/l);
5  beta = 1;
6  gamma = 10;
7  omega0 = 1;
8
9  Abf = poly([-alpha; -alpha; -beta; -gamma; ...
10             -beta-1i*omega0; -beta+1i*omega0]);
11
12  c1 = Abf(2);
13  c2 = Abf(3);
14  c3 = Abf(4);
15  c4 = Abf(5);
16  c5 = Abf(6);
17  c6 = Abf(7);
18
19  fprintf(['c1=%.4f c2=%.4f c3=%.4f c4=%.4f ' ...
20          'c5=%.4f c6=%.4f\n'], c1,c2,c3,c4,c5,c6)

```

Listing A.1: Closed-loop pole assignment and polynomial coefficients.

A.2 Computation of $R(s)$ and $S(s)$

```

1  %% R and S polynomials
2  sigma1 = c1;
3  r0 = (omega0^2-alpha^2-c2)/alpha^2;
4  r1 = (c1*(omega0^2-alpha^2)-c3)/alpha^2;
5  r2 = -(alpha^2*omega0^2+c4)/alpha^2;
6  r3 = -(alpha^2*c1*omega0^2+c5)/alpha^2;
7  r4 = -c6/alpha^2;
8
9  fprintf('sigma1=%.4f\n', sigma1)
10 fprintf('r0=%.4f r1=%.4f r2=%.4f r3=%.4f r4=%.4f\n', ...
11         r0,r1,r2,r3,r4)

```

Listing A.2: Bézout-equation coefficient identification.

A.3 Computation of $T(s)$

```

1 %% T polynomial
2 t2 = r4/(gamma*beta);
3 M = [1-gamma*beta, -(gamma+beta); ...
4      -(gamma+beta), gamma*beta-1];
5 Y = [r0-r2+r4-(gamma*beta-1)*t2; ...
6      r3-r1-(gamma+beta)*t2];
7 Sol = M\Y;
8 t0 = Sol(1);
9 t1 = Sol(2);
10
11 fprintf('t0=%.4f t1=%.4f t2=%.4f\n', t0,t1,t2)

```

Listing A.3: Feedforward-polynomial calculation.

A.4 Stability indicators

```

1 %% Controller analysis
2 R_o = [r0_o, r1_o, r2_o, r3_o, r4_o];
3 S_o = conv([1 0 1 0], [1 sigma1_o]);
4 T_o = conv([1 beta_opt+gamma_opt beta_opt*gamma_opt], ...
5            [t0_o t1_o t2_o]);
6 A_o = [1, 0, -alpha^2];
7 B_o = -alpha^2;
8 num_B0_o = conv(B_o, R_o);
9 den_B0_o = conv(S_o, A_o);
10
11 [Gm_o, Pm_o, Wcg_o, Wcp_o] = margin(num_B0_o, den_B0_o);
12 [re_o, im_o, ~] = nyquist(tf(num_B0_o, den_B0_o));
13 re_o = squeeze(re_o);
14 im_o = squeeze(im_o);
15 Mm_o = min(sqrt((re_o+1).^2 + im_o.^2));
16
17 fprintf('\nStability margins:\n')
18 fprintf(' Gain margin: %.4f dB\n', 20*log10(Gm_o))
19 fprintf(' Phase margin: %.4f deg\n', Pm_o)
20 fprintf(' Modulus margin: %.4f\n', Mm_o)
21 fprintf(' Gain crossover frequency: %.4f rad/s\n', Wcg_o)
22 fprintf(' Phase crossover frequency: %.4f rad/s\n', Wcp_o)

```

Listing A.4: Frequency-domain stability calculations.

A.5 Optimized-controller plots and Simulink simulation

```

1 %% Frequency plots - optimized RST
2 figure('Name','Optimized RST Bode','NumberTitle','off');
3 margin(num_B0_o, den_B0_o);
4 title('Open-loop Bode diagram - optimized RST (beta=0.5, gamma=30)')
5 grid on;
6
7 figure('Name','Optimized RST Nyquist','NumberTitle','off');
8 nyquist(num_B0_o, den_B0_o);
9 title('Open-loop Nyquist diagram - optimized RST (beta=0.5, gamma=30)')

```

```

10 axis([-2 0.5 -2 2]); grid on; hold on;
11 theta_circle = linspace(0,2*pi,200);
12 plot(-1 + Mm_o*cos(theta_circle), Mm_o*sin(theta_circle), ...
13      'r--', 'LineWidth', 1.5)
14 plot(-1,0, 'r+', 'MarkerSize', 12, 'LineWidth', 2)
15 legend('Open-loop Nyquist', ...
16        sprintf('Modulus-margin circle M_m = %.3f', Mm_o), ...
17        'Critical point -1', 'Location', 'best')
18 hold off;
19
20 %% Simulink simulation - optimized RST
21 beta = beta_opt;
22 gamma = gamma_opt;
23 sigma1 = sigma1_o;
24 r0 = r0_o; r1 = r1_o; r2 = r2_o; r3 = r3_o; r4 = r4_o;
25 t0 = t0_o; t1 = t1_o; t2 = t2_o;
26 R = R_o; S = S_o; T = T_o;
27 H = poly([-2 -2 -1 -1]);
28 sim('simu_rst_pendule')
29
30 figure('Name','Optimized RST time response','NumberTitle','off');
31 subplot(4,1,1)
32 plot(t,ymc,'g','LineWidth',1.5); hold on
33 plot(t,ym,'r','LineWidth',1.2)
34 legend('Reference y_{mc}','Output y_m','Location','best')
35 xlabel('Time (s)'); ylabel('Position (V)')
36 title('Reference tracking - optimized RST (beta=0.5, gamma=30)')
37 ylim([-3 3]); grid on
38
39 subplot(4,1,2)
40 plot(t,ym-ymc,'c','LineWidth',1.2)
41 xlabel('Time (s)'); ylabel('Error (V)')
42 title('Tracking error e(t) = y_m(t)-y_{mc}(t)'); grid on
43
44 subplot(4,1,3)
45 plot(t,u,'m','LineWidth',1.2); hold on
46 yline(5,'r--','+5 V limit'); yline(-5,'r--','-5 V limit')
47 xlabel('Time (s)'); ylabel('Command u (V)')
48 title('Control command u(t)'); ylim([-7 7]); grid on
49
50 subplot(4,1,4)
51 plot(t,Theta,'Color',[1 0.5 0],'LineWidth',1.2)
52 xlabel('Time (s)'); ylabel('Angle theta (deg)')
53 title('Pendulum angle theta(t)'); grid on

```

Listing A.5: Optimized frequency- and time-domain evaluation.

A.6 Robustness test for pendulum-length variation

```

1 %% Robustness test: pendulum-length variation
2 L_values = [0.16, 0.23, 0.28, 0.35];
3 colors = {'r','b','m','c'};
4 l_original = l;
5 results_t = cell(length(L_values),1);
6 results_ymc = cell(length(L_values),1);
7 results_ym = cell(length(L_values),1);
8 results_u = cell(length(L_values),1);
9 results_th = cell(length(L_values),1);
10

```

```

11 fprintf('\n===== ROBUSTNESS TEST - Variation of L =====\n')
12 fprintf('(Fixed optimized RST: beta=%.2f, gamma=%.2f)\n\n', ...
13         beta_opt,gamma_opt)
14 fprintf('%-10s %-10s %-15s %-15s %-15s %-12s\n', ...
15         'L (m)', 'alpha', 'Gm (dB)', 'Pm (deg)', 'Mm', '|u|max (V)')
16
17 for k = 1:length(L_values)
18     l = L_values(k);
19     alpha_k = sqrt(g/l);
20     A_k = [1,0,-alpha_k^2];
21     B_k = -alpha_k^2;
22     num_B0_k = conv(B_k,R_o);
23     den_B0_k = conv(S_o,A_k);
24     [Gm_k,Pm_k,~,~] = margin(num_B0_k,den_B0_k);
25     [re_k,im_k,~] = nyquist(tf(num_B0_k,den_B0_k));
26     re_k = squeeze(re_k); im_k = squeeze(im_k);
27     Mm_k = min(sqrt((re_k+1).^2 + im_k.^2));
28     sim('simu_rst_pendule')
29     results_t{k} = t;
30     results_ymc{k} = ymc;
31     results_ym{k} = ym;
32     results_u{k} = u;
33     results_th{k} = Theta;
34     fprintf('%-10.2f %-10.4f %-15.4f %-15.4f %-15.4f %-12.4f\n', ...
35             l,alpha_k,20*log10(Gm_k),Pm_k,Mm_k,max(abs(u)))
36 end
37 l = l_original;
38
39 labels_ym = arrayfun(@(L) sprintf('y_m (L=%.2f m)',L), ...
40                     L_values,'UniformOutput',false);
41 labels_th = arrayfun(@(L) sprintf('theta (L=%.2f m)',L), ...
42                     L_values,'UniformOutput',false);
43 labels_u = arrayfun(@(L) sprintf('u (L=%.2f m)',L), ...
44                     L_values,'UniformOutput',false);
45
46 figure('Name','Robustness test - variation of L','NumberTitle','off')
47 subplot(3,1,1)
48 plot(results_t{1},results_ymc{1},'g','LineWidth',1.8); hold on
49 for k=1:length(L_values)
50     plot(results_t{k},results_ym{k},colors{k},'LineWidth',1.2)
51 end
52 legend(['Reference y_{mc}'],labels_ym,'Location','best')
53 xlabel('Time (s)'); ylabel('Position (V)')
54 title('Reference tracking - robustness to variation of L')
55 ylim([-3 3]); grid on
56
57 subplot(3,1,2); hold on
58 for k=1:length(L_values)
59     plot(results_t{k},results_th{k},colors{k},'LineWidth',1.2)
60 end
61 legend(labels_th,'Location','best')
62 xlabel('Time (s)'); ylabel('Angle theta (deg)')
63 title('Pendulum angle theta(t)'); grid on
64
65 subplot(3,1,3); hold on
66 for k=1:length(L_values)
67     plot(results_t{k},results_u{k},colors{k},'LineWidth',1.2)
68 end
69 yline(5,'k--','+5 V limit'); yline(-5,'k--','-5 V limit')
70 legend(labels_u,'Location','best')
71 xlabel('Time (s)'); ylabel('Command u (V)')
72 title('Control command u(t)'); ylim([-7 7]); grid on

```

Listing A.6: Parametric robustness analysis.

Project Deliverables

The original project package associated with this report contains:

- **MATLAB code:** TP2_RST_Pendule_Dev_KUMAR.m;
- **Simulink model:** simu_rst_pendule.mdl;
- **Python pole-placement visualization:** placement_poles.py.

END OF REPORT